Generative Sketching for Relational Joins

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ABOUT ME

Research interests

Generative Models → Create Sketches → Faster Databases

Studied generative models to create sketches for better database query optimization:

- Tsan et al., “Approximate Sketches,” SIGMOD 2024 — transformers.
- Tsan et al., “Sketched Sum-Product Networks for Joins,” VLDB 2027 — sum-product networks.

OVERVIEW

What we'll cover

01 What is sketching?

join size · inner products · embedding distributions

02 Tug-of-War Sketch

03 **Generative Sketching — my work**

04 Results: 8.6× faster than PostgreSQL on JOB

01

SECTION

What is sketching?

Sublinear summaries, and three examples.

1 · DEFINITION

What is sketching?

The usage of “sketch” is somewhat colloquial. It refers to data structures that summarize data in sublinear space — usually in a single pass. Sometimes it refers to random projection, called linear sketching.

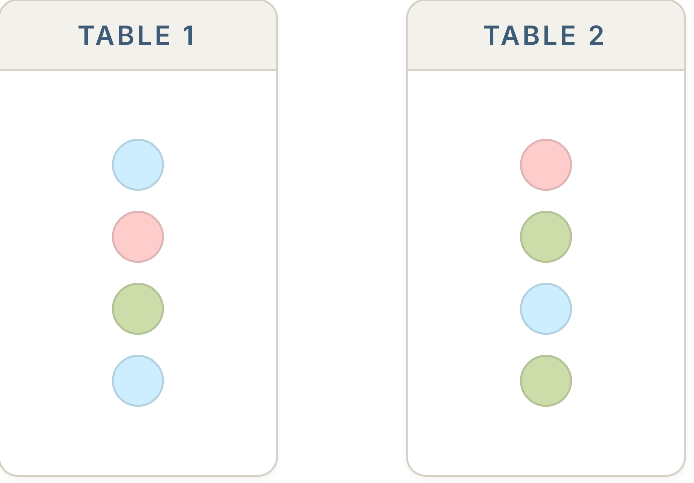
EXAMPLE APPLICATIONS

- **Join sizes** — used in relational database systems (our focus)
- **Inner products**
- **Embedding distributions**

1 · DEFINITION

Join size

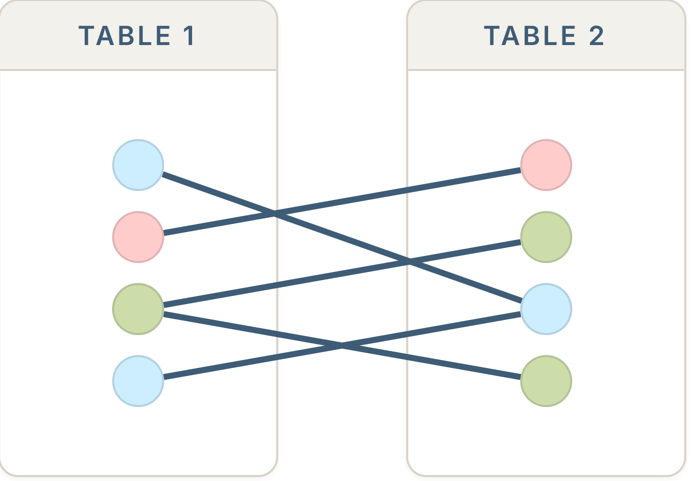
How many rows match?



1 · DEFINITION

Nested-loop join

Compare every Table 1 row against every Table 2 row; a line marks a match.

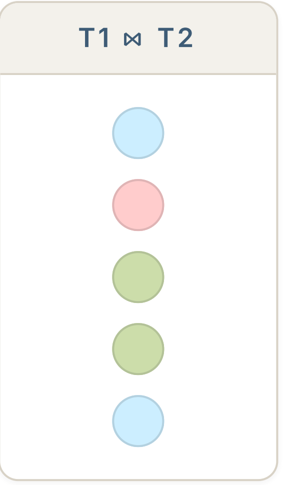


$$|T_1 \bowtie T_2| = 5$$

1 · DEFINITION

The join result

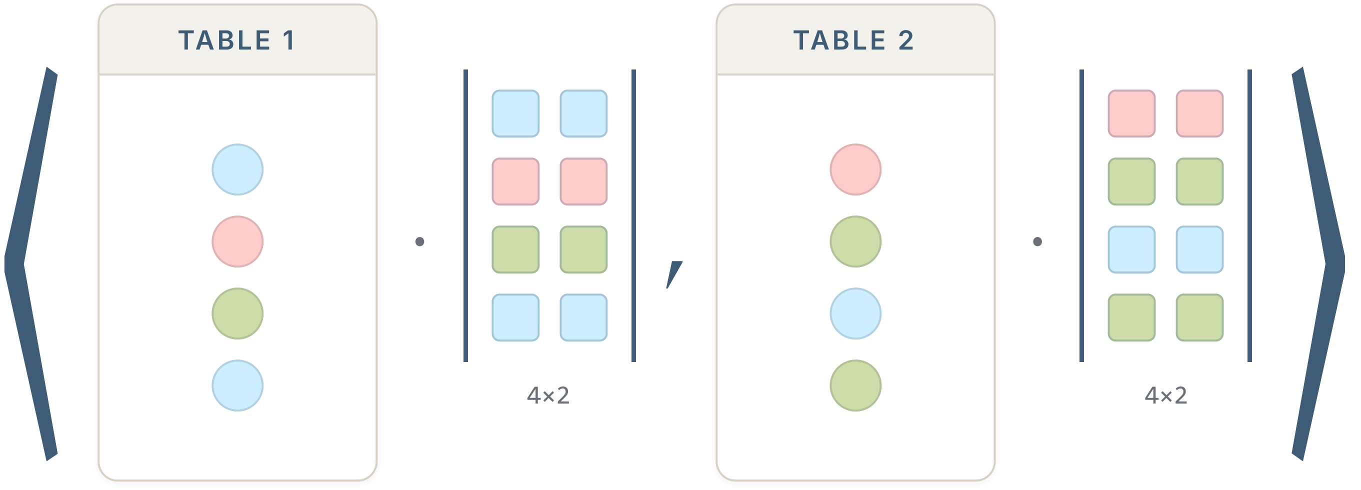
Each match merges into one result row.



1 · DEFINITION

Join size sketching

Project each table through the same random values per key — 4 rows down to 2.

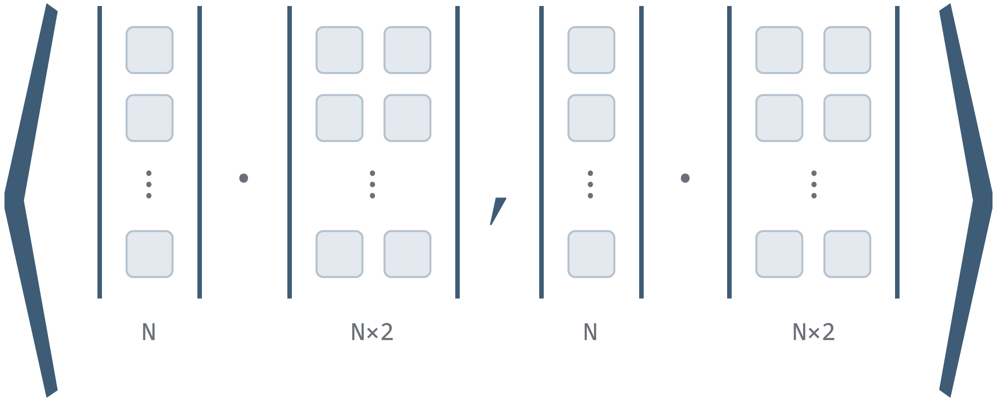


This inner product approximates the join size $|T_1 \bowtie T_2|$.

1 · DEFINITION

Inner product sketching

Project each N-dimensional vector through the same sketching matrix.

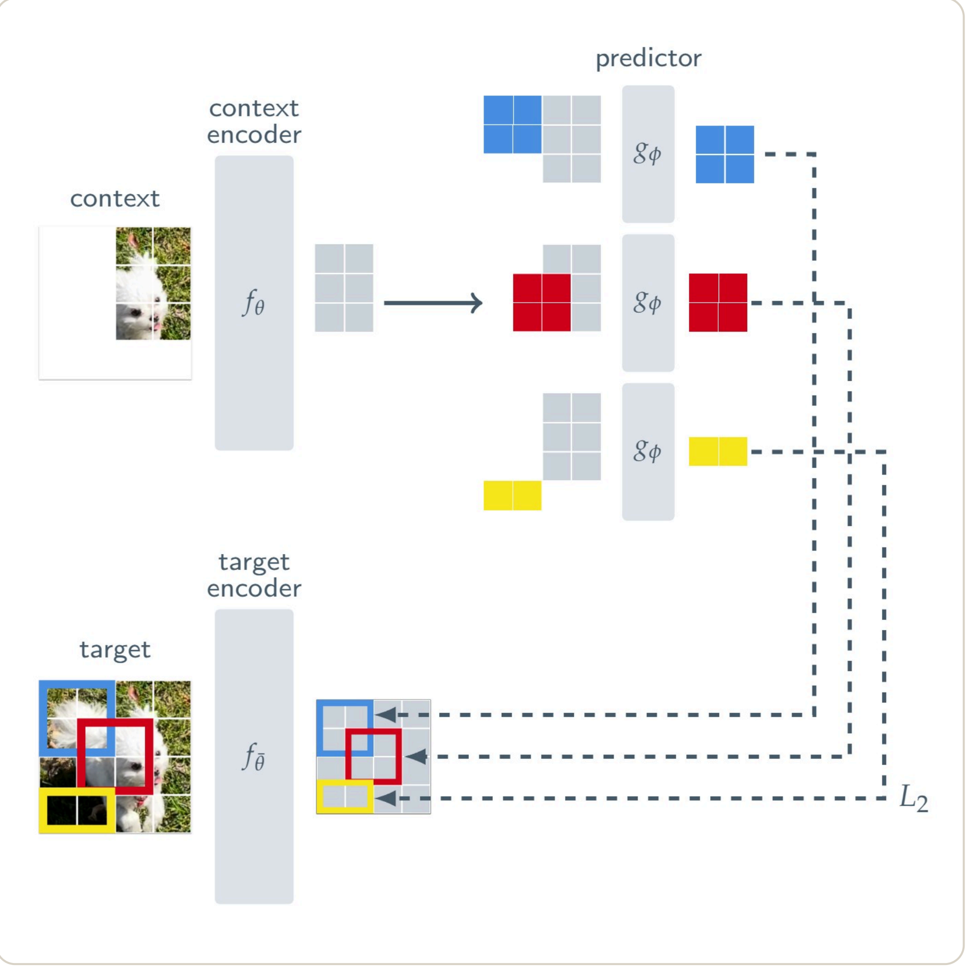


$$\langle S^\top x_1,\, S^\top x_2 \rangle \approx \langle x_1, x_2 \rangle$$

1 · DEFINITION

Embedding distribution

A context x and a target y are encoded into embeddings s_x, s_y; a predictor takes s_x and a condition z and predicts s_y. Having to predict is what makes the encoder learn richer, information-preserving embeddings.



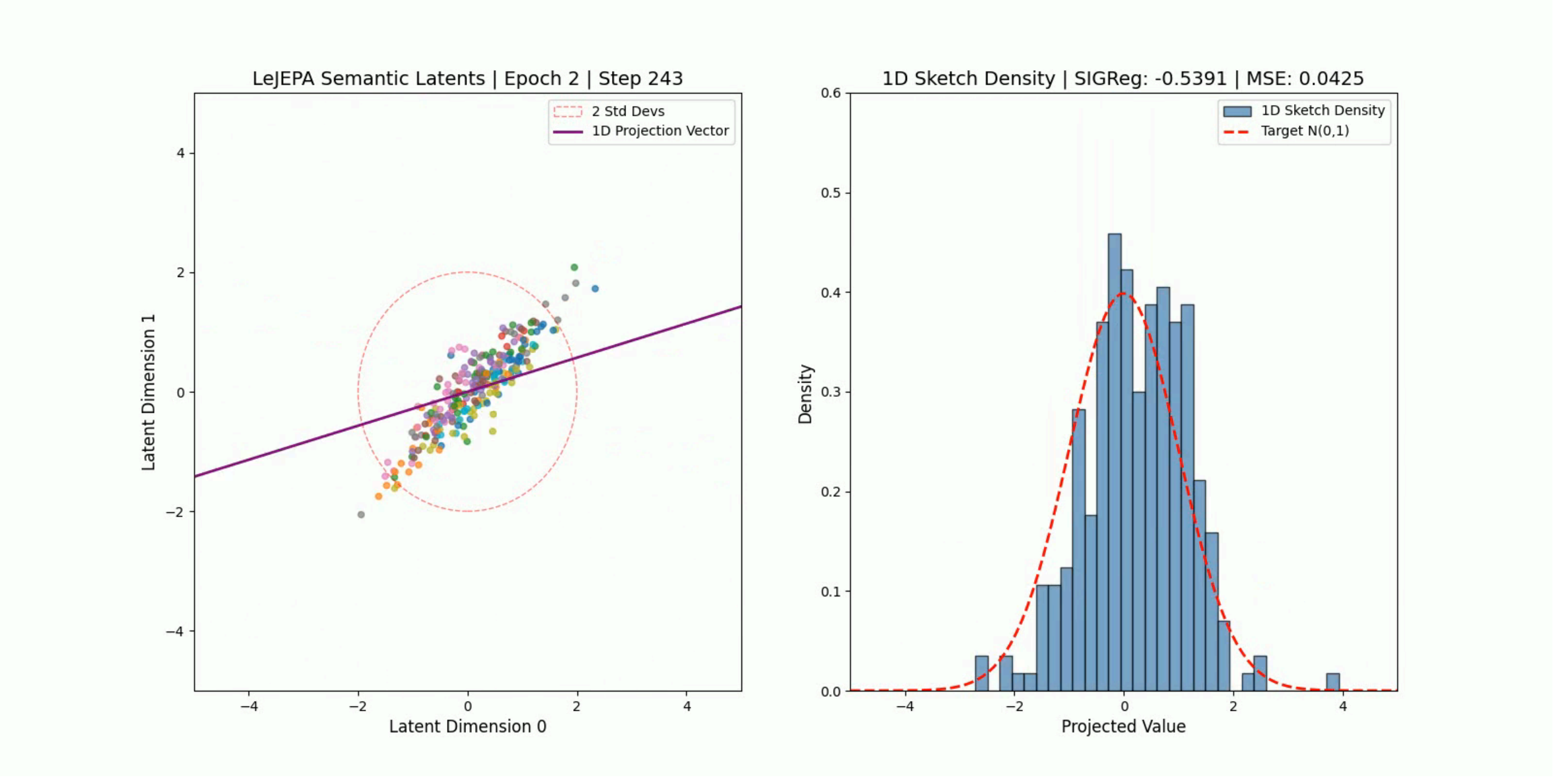
Assran et al., "Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture," CVPR 2023.

github.com/facebookresearch/ijepa

1 · DEFINITION

Sketched Isotropic Gaussian Regularization

The CIFAR-10 embedding distribution as it trains.



A random 1-D projection of the latents (left), and the density of that sketch (right)

1 · DEFINITION

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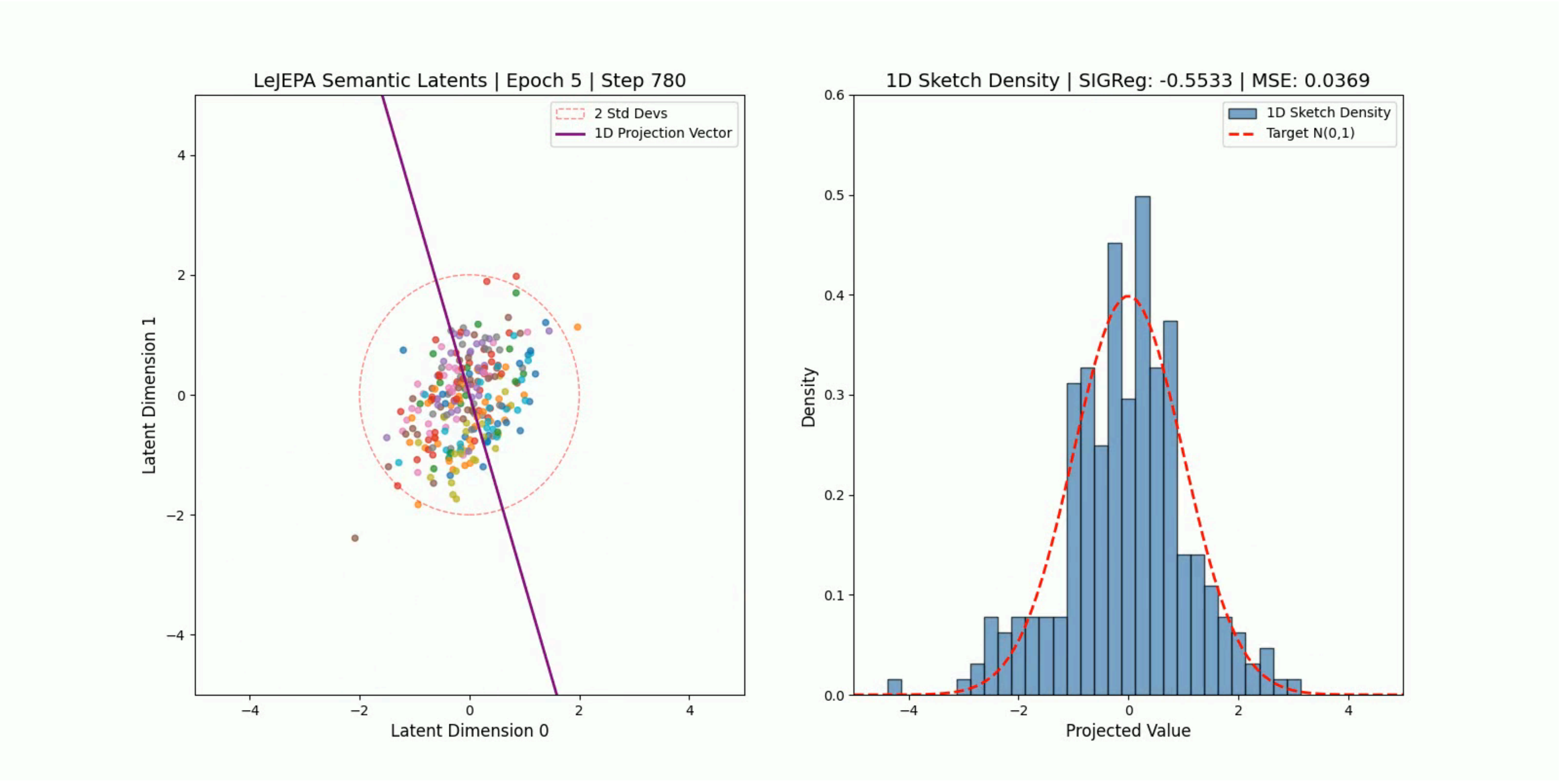
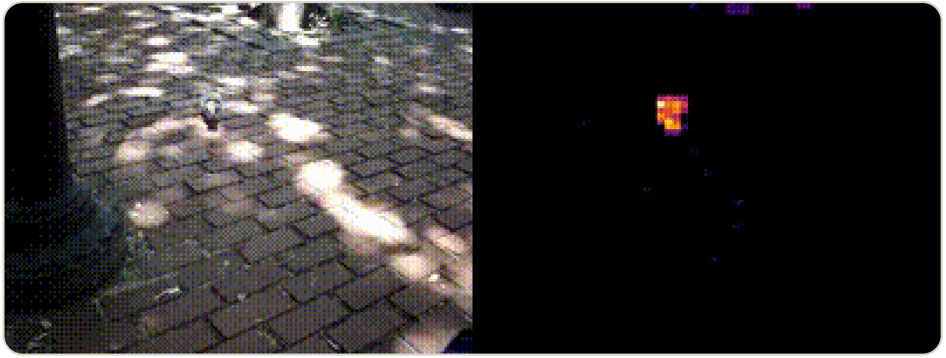


A random 1-D projection of the latents (left), and the density of that sketch (right)

1 · DEFINITION

Results from LeJEPA

Plain video (left), attention map demonstrating LeJEPA's emergent salient object tracking behavior (right).



Balestriero and LeCun, "LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics," arXiv 2025.

github.com/galilai-group/lejepa

02

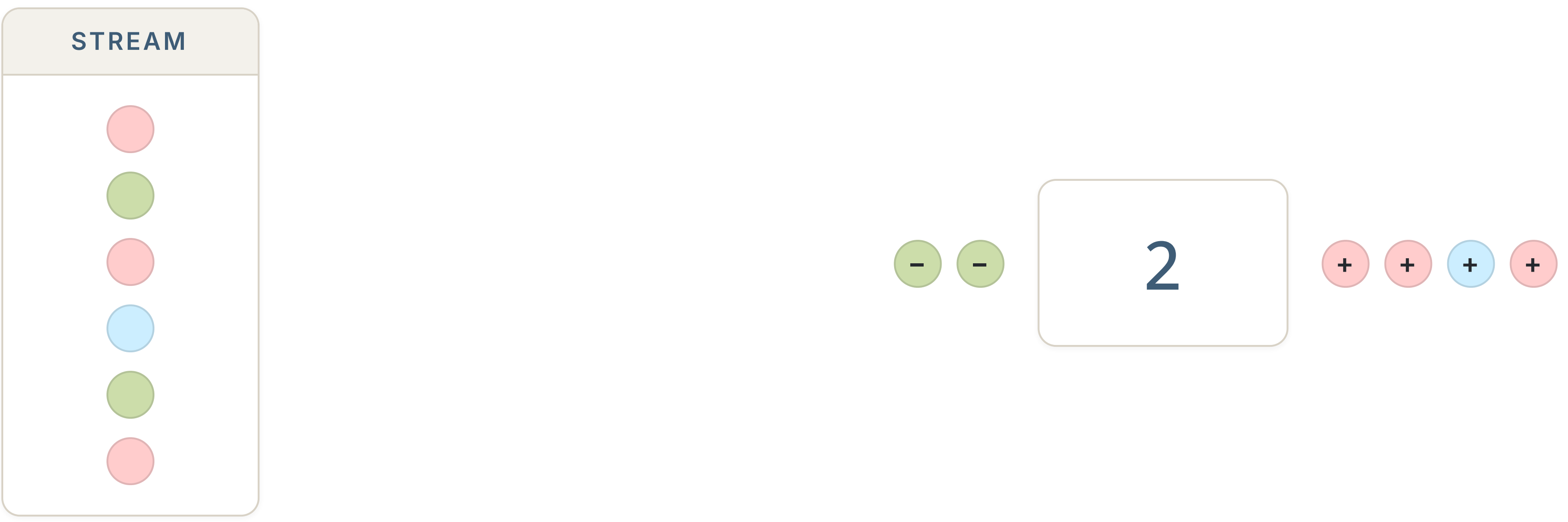
SECTION

Tug-of-War Sketch

Alon, Matias, and Szegedy, “The space complexity of approximating the frequency moments,” STOC 1996.

Frequency estimation

Every item pulls the counter one way or the other.

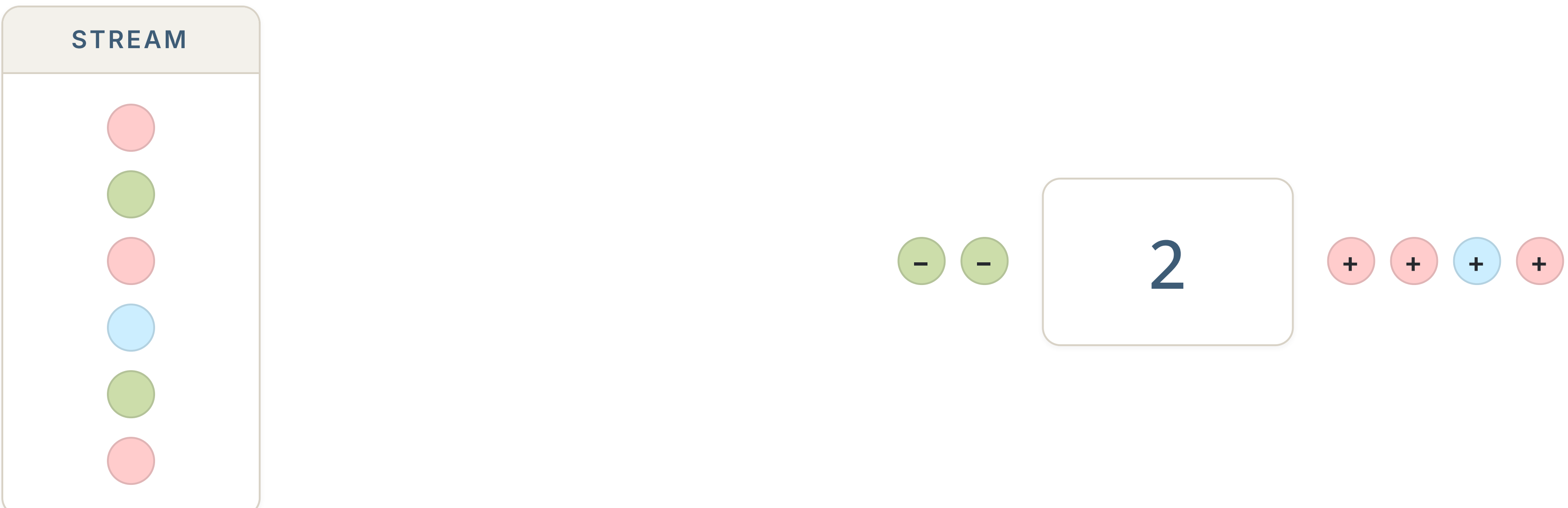


Each item pulls the counter one way.

$$Z = \sum_i s(x_i) = 2$$

Estimating each count

Multiply the counter by that colour's own sign.



$(+1) \cdot 2 = \mathbf{2}$

true 3

$(-1) \cdot 2 = \mathbf{-2}$

true 2

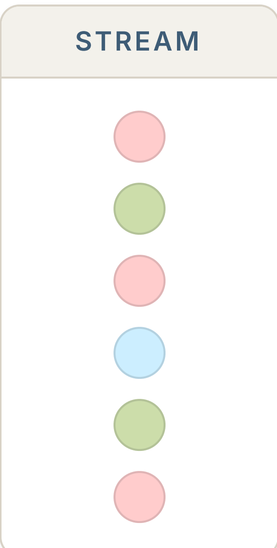
$(+1) \cdot 2 = \mathbf{2}$

true 1

Was it correct?

Averaging many games

Re-draw the signs each round; every round is an independent estimate.



$$\xi : [N] \to \{-1, +1\}$$

– – – – – **-4** +

$$\mathbb{E}\left[\xi(i)\,\xi(j)\right] = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

avg **3.69**	avg **2.77**	avg **1.85**
true 3	true 2	true 1

round **13**

Join size estimation

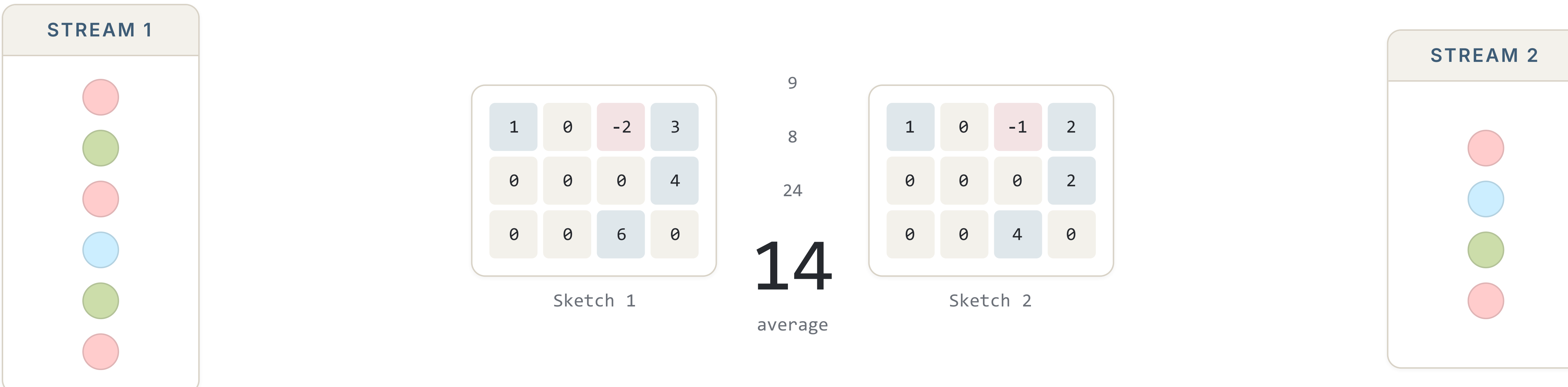
Two streams, one counter each — sketched with the *same* signs ξ.

TRUE JOIN SIZE

$(3 \times 2) + (2 \times 1) + (1 \times 1) = \mathbf{9}$

Count sketch

Width splits collisions into buckets; depth adds independent rows, combined by median.

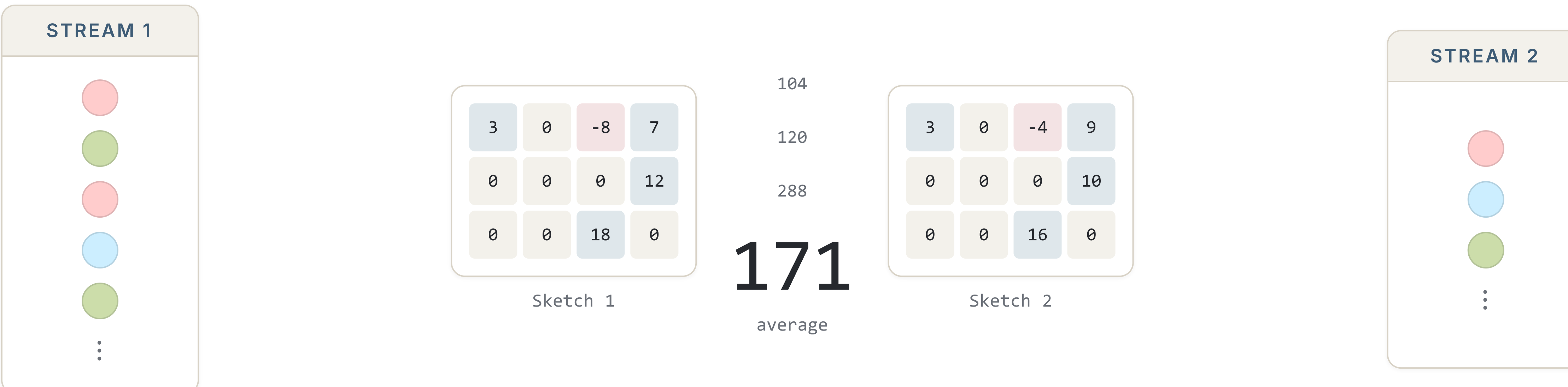


WIDTH 4 DEPTH 3

$(3 \times 2) + (2 \times 1) + (1 \times 1) = \mathbf{9}$

A live stream

The stream keeps growing; each arrival lands in one bucket per row.



WIDTH 4 DEPTH 3

$(7 \times 9) + (8 \times 4) + (3 \times 3) = \mathbf{104}$

03

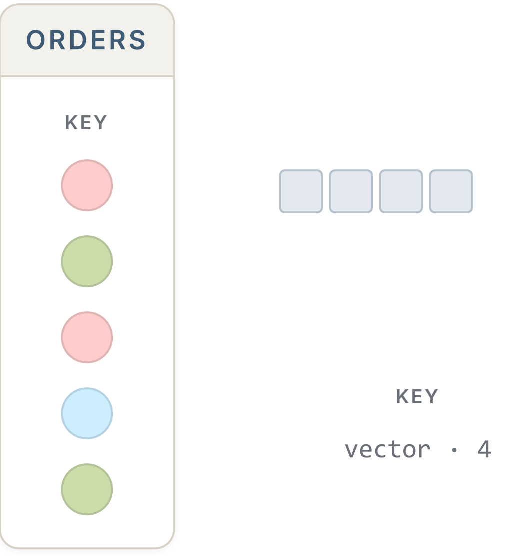
SECTION

Generative Sketching

Learning a joint distribution to generate sketches.

The problem with databases

Tables usually have more than just one column.



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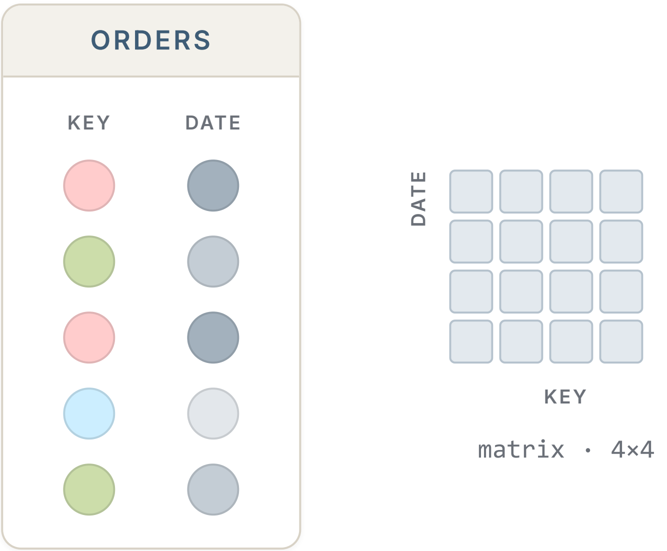

3 · GENERATIVE SKETCHING

The problem with databases

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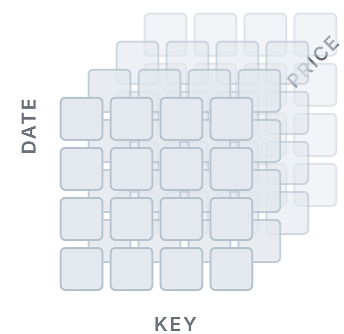
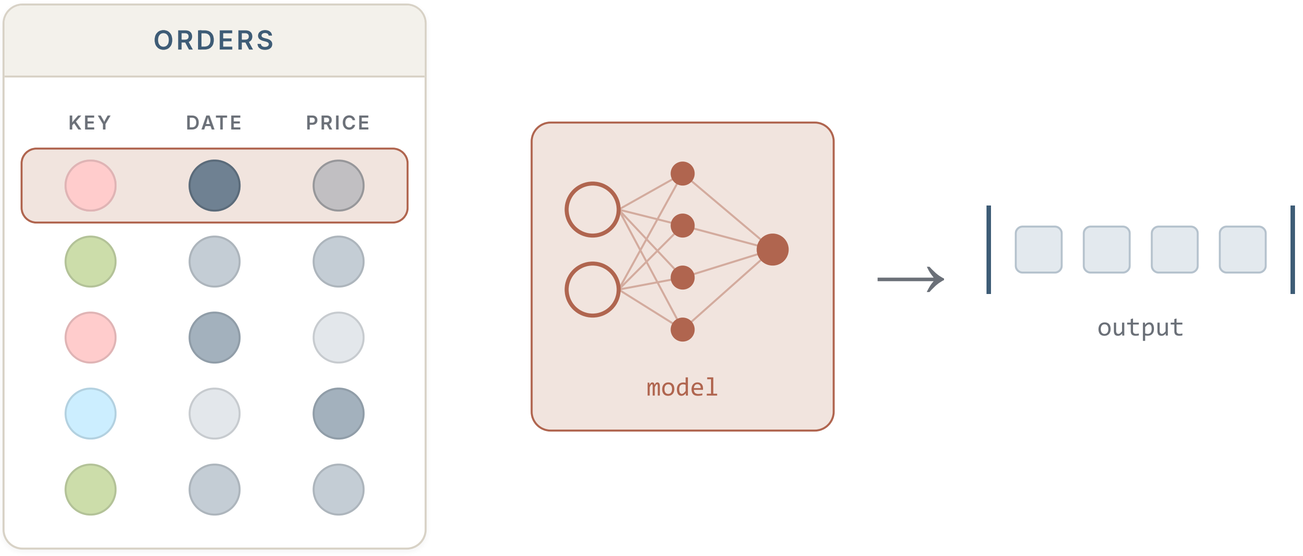


tensor · 4×4×4

3 · GENERATIVE SKETCHING

Training the model

Every row is an example: the other columns in, the join key out.



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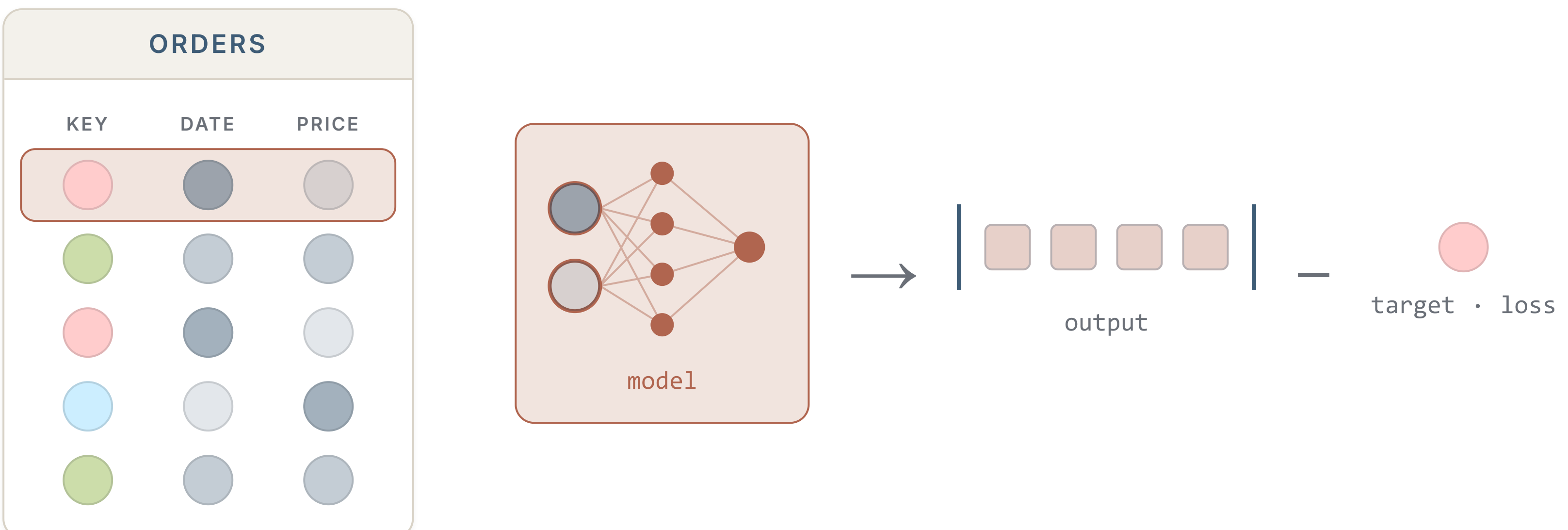
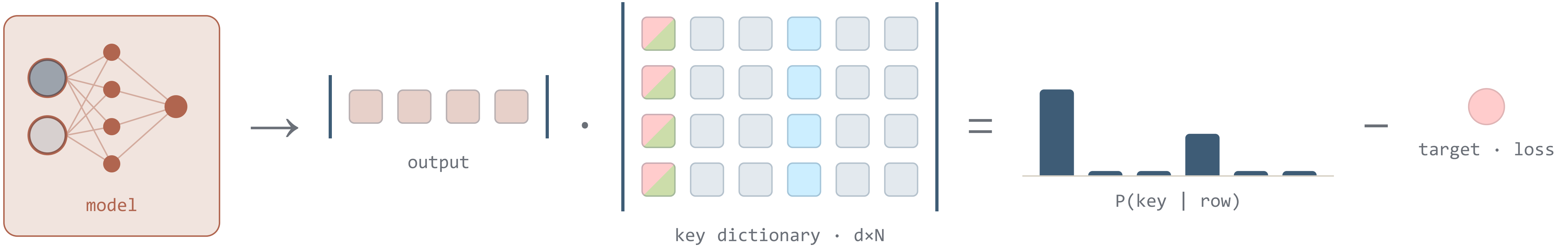


Learning the key embeddings

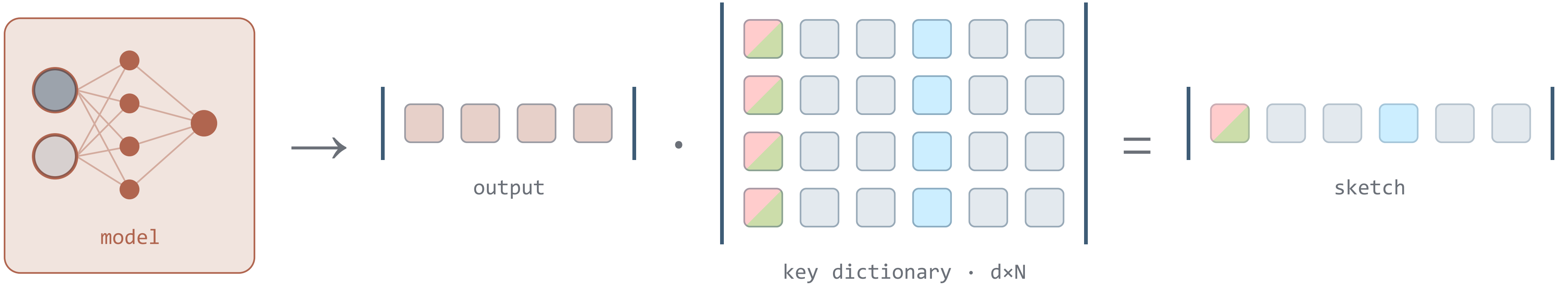
Softmax over one embedding per key: the dictionary is an output layer of the model, trained with it.



3 · GENERATIVE SKETCHING

Generating the sketch

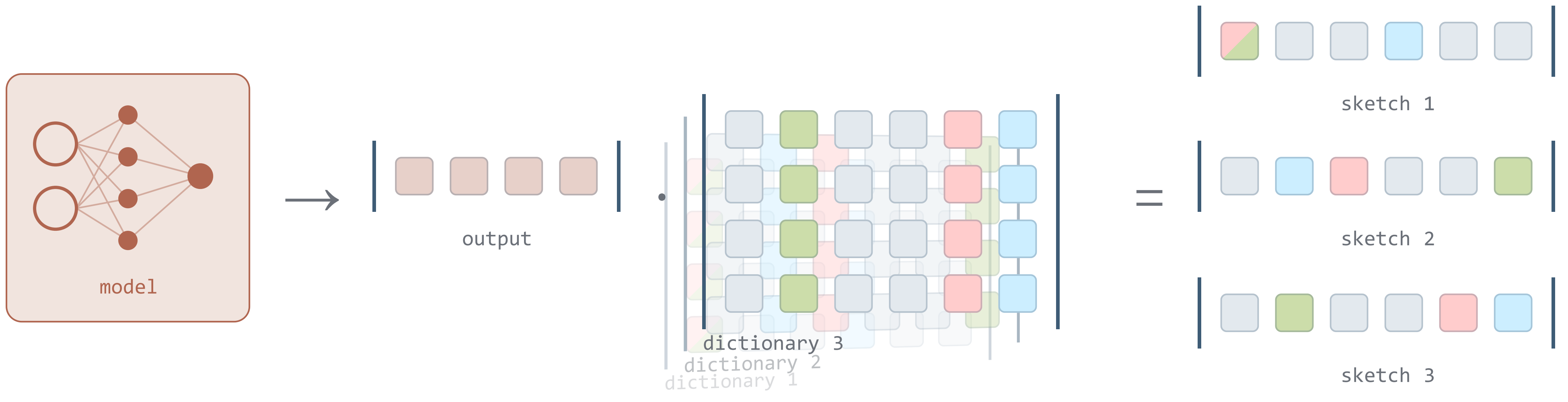
The same product the loss trained — one dot product against the key embeddings gives the whole sketch.



3 · GENERATIVE SKETCHING

Independent sketches

Each sketch has its own embedding dictionary.



04

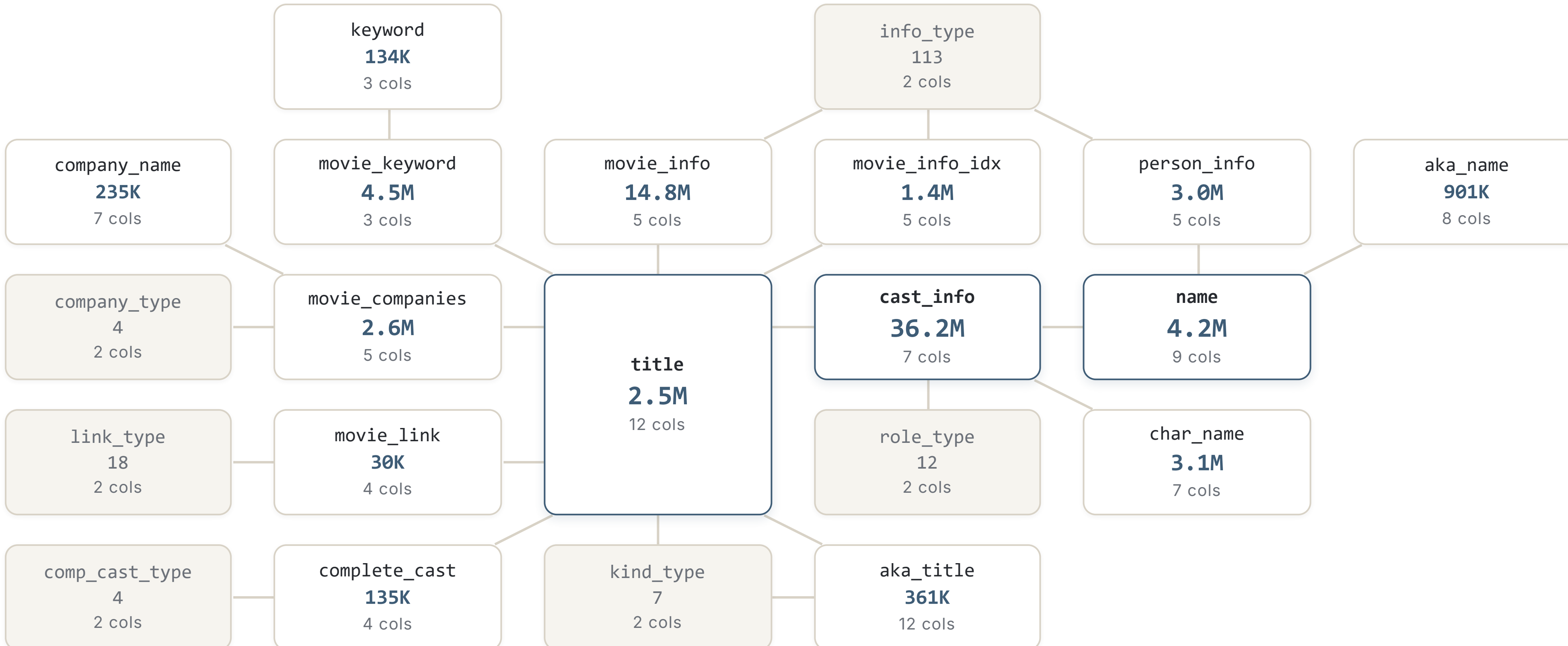
SECTION

8.6× faster than PostgreSQL

113 Join Order Benchmark queries against PostgreSQL 18.4.

4 · RESULTS

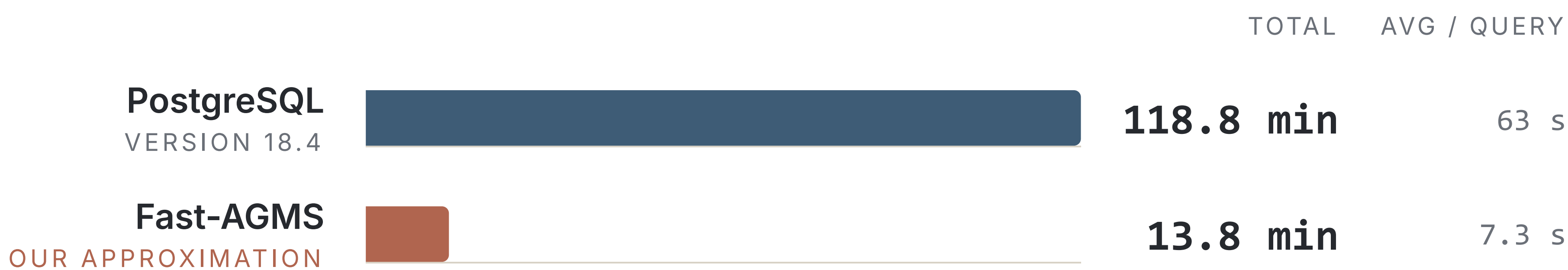
The Join Order Benchmark



21 tables · 108 columns · 74,190,187 rows · 23 join edges

4 · RESULTS

Total query execution time



8.6× faster than PostgreSQL

113 Join Order Benchmark queries · IMDb, 74,190,187 rows
Tsan et al., “Sketched Sum-Product Networks for Joins,” VLDB 2027.

WRAP-UP

Conclusion

01 Sketching has various applications — join size, inner products, embedding distributions, etc.

02 **Generative models enable sketching for multivariate data.**

03 *This talk: 10% content, 90% slide animations.*

BONUS SLIDE

Vector–matrix–vector product

One sketch per factor: x and y by their own hashes, A by the circular convolution of both — bucket $(h_1(i) + h_2(j)) \bmod w$, sign $\xi_1(i)\xi_2(j)$.

			ξ₂=−1	ξ₂=−1	ξ₂=−1
		y	3	1	1
	x		h₂=0	h₂=0	h₂=0
ξ₁=−1	3	h₁=2	3	1	1
ξ₁=−1	2	h₁=1	0	3	2
ξ₁=+1	4	h₁=1	3	3	1
ξ₁=+1	2	h₁=1	3	1	3

xTAy = **121**

Sx	0	4	-3	0
SA	0	-9	5	0

★ CIRCULAR CROSS-CORRELATION

Sx⋆SA	-51	20	0	27

· DOT

Sy	-5	0	0	0

255

avg **219.46** over 13 draws true **121**

WIDTH 4